CCT Bakery: A Java GUI Application Using Queue Data Structure

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Lecture: Algorithms and Constructs | **Lecturer:** Dr Muhammad Iqbal | **Institution:** CCT College Dublin | **Date**: 18th April 2025

This is an individual CA1 assignment for Algorithms and Constructs. You are required to solve a problem using JAVA programming language based on the below mentioned scenario:

A confectioner provides five different types of cakes for parties: Pineapple, Strawberry, Chocolate, Vanilla, and Plain cake. These cakes are baked in a rectangular oven with two doors: one at the front and one on the opposite side. However, the baker can either use the front door for both adding and removing cakes or use the front door for adding cakes and the opposite door for removing them. The oven can hold up to five cake pans at a time, and the cakes have different weights and names that must be tracked as they are placed into and removed from the oven. The Java application should store each cake's information, including its name, weight (grams), the best-before date (maximum of two weeks), and the time it was placed into the oven. For example, (name: Plain cake, weight: 200 grams).

Design and develop a Java application that models this scenario using an appropriate data structure (Queue or Stack or any other of your choice). The application must have a GUI that supports the following operations.

The Java application stores and manipulates data representing cakes. The Cakes class stores the information of the cakes such as type, weight, date of birth, and best before date. Users can insert cakes into the oven with a maximum of 5 cakes, can view the cakes in the oven, move them from the oven to the shelf using Queue data structure, and remove them.

QUEUE: PROPERTIES

A **queue** is the most appropriate data structure for a bakery oven because it follows a **First-In, First-Out (FIFO)** approach, allowing the cakes being baked and removed in the order that they were added, imitating a real-world operations. This data structure also supports operations like adding, removing, and displaying which allows users to move the cakes between the oven and the shelf.

PROPERTIES

- Queues are FIFO data structures. All insertions are done at the rear and all deletions at the front.
- The front always refers to the first element in the queue.
- The rear always refers to the position where the next element is to be inserted in the queue.
- The queue has a fixed upper bound (capacity) on the number of elements it can store.

SYSTEM DESIGN

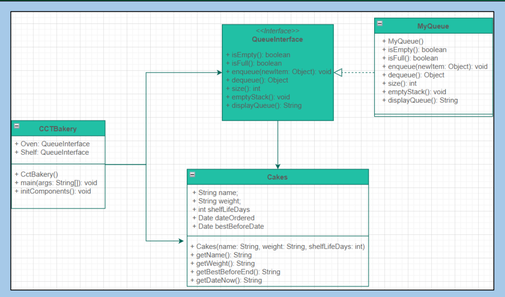
JAVA IMPLEMENTATION

- ✓ **QueueInterface.java:** Interface defining queue operations
- ✓ **MyQueue.java:** Queue logic implementation
- ✓ **CctBakery.java:** GUI and user interaction logic
- ✓ **Cakes.java:** Data structure to store name, weight, best-before, and Best Before End.

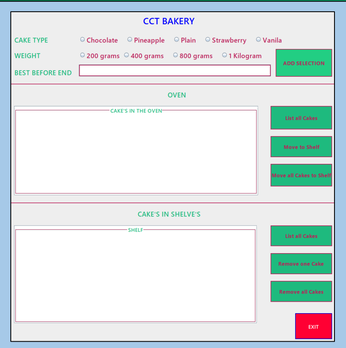
OPERATIONS

- **enqueue():** Inserts a cake at the rear of the queue.
- **dequeue():** Removes a cake from the front of the queue and return it; a popup message will be displayed indicating that the queue is empty if the queue is empty.
- **size():** Returns the number of objects in the queue.
- **isEmpty():** Returns a Boolean value that indicates whether the queue is empty.

Unified Modelling Language (UML) Diagram



CCT BAKERY: GUI APPLICATION



USER INPUT OPTIONS:

- ✓ **Cake Type:** Choose one between 5 preset options.
- ✓ **Weight:** Select one between 4 preset weights.
- ✓ **Best Before End:** Enter shelf life in days (displayed as a date in the format dd-MM-YYYY).

CAKE ADDITION:

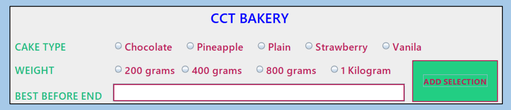
- ✓ Use the ADD SELECTION button to insert a cake into the oven.
- ✓ The app displays clear messages when:
- Required fields are missing or incorrect. A cake is successfully added.
- The oven reaches its maximum capacity of 5 cakes.

CAKE MANAGEMENT CONTROLS:

- ✓ ***Move to Shelf:*** Transfers the first cake from the oven to the shelf (FIFO).
- ✓ ***Move All Cakes to Shelf:*** Transfers all cakes from the oven to the shelf. Shelf capacity is equal to the oven.
- ✓ **List Cakes:** Displays all cakes currently stored in the shelf.
- ✓ **Remove One Cake:** Remove the first cake added to the shelf.
- ✓ **Remove All Cakes:** Remove all cakes from the shelf at once.
- ✓ **Exit:** Closes the application.

CCT BAKERY: CAKE SELECTION

In this section, the app allows the user to select the **Cake Type** by the selection of it respective *RadioButton*, same logic for the **Weight** selection.



The **ADD SELECTION** button handles the different options and also popup with a proper message depending if one of the field is not selected or well typed as is the case of the Best Before End.

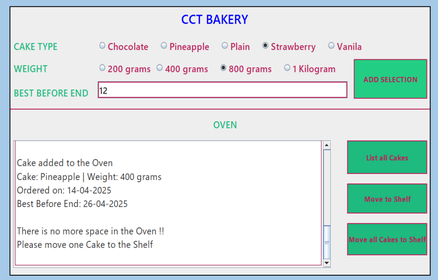






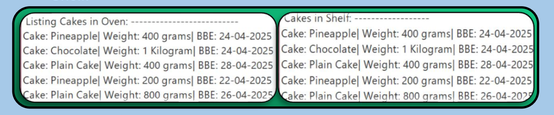
Once the cake options are fulfilled and the **ADD SELECTION** button is clicked, the cake will be displayed in the Oven displaying a message:

- ✓ The information of the cake added.
- ✓ Date Ordered and Best Before End, dates which were calculated using the **getDateNow()** and **getBestBeforeEnd()** methods from the Cake class.
- ✓ Once 5 cakes were added to the oven, this will display a message indicating that the oven is full.



CCT BAKERY: BUTTONS & FUNNCTIONALITY

- **List all Cakes:** Displays the cakes that are stored in the Oven and Shelf.



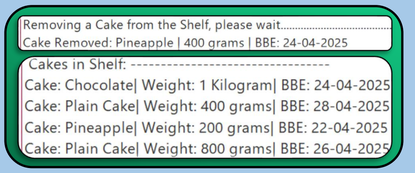
- **Move to Shelf button:** Displays a message indicating that the first Cake added to the Oven were moved to the Shelf



- **Move all Cakes to Shelf button:** Move all Cakes from the Oven to the Shelf displaying a message with the information of the cakes moved.



- **Remove one Cake button:** Remove the fist cake added to the Shelf and displays a message indicating that a cake is being removed with their respective information will be



- **Remove all Cakes** buttons: remove all the cakes from the Shelf, displaying a message indicating that the Shelf is empty.



- **Exit button:** Stops and close the app at the moment, and was coded as it follows:

```
//Exit button
private void jButton7ActionPerformed(java.awt.event.ActionEvent evt) {
    System.exit(0);
}
```

REFERENCES

Iqbal, M., 2025. Lecture 5: Algorithms and Constructs – Queue Implementation. [slides] Week 5, HDip. in Science in Computing. CCT College Dublin. Slides 5, 7, 9, 10, 11.

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